



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

50

Locations

Key features span **demographics** (Age, Gender, Location, Subscription), **purchase details** (Item, Category, Amount, Season), and **shopping behavior** (Discount, Promo Code, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

Cleaning & Standardization

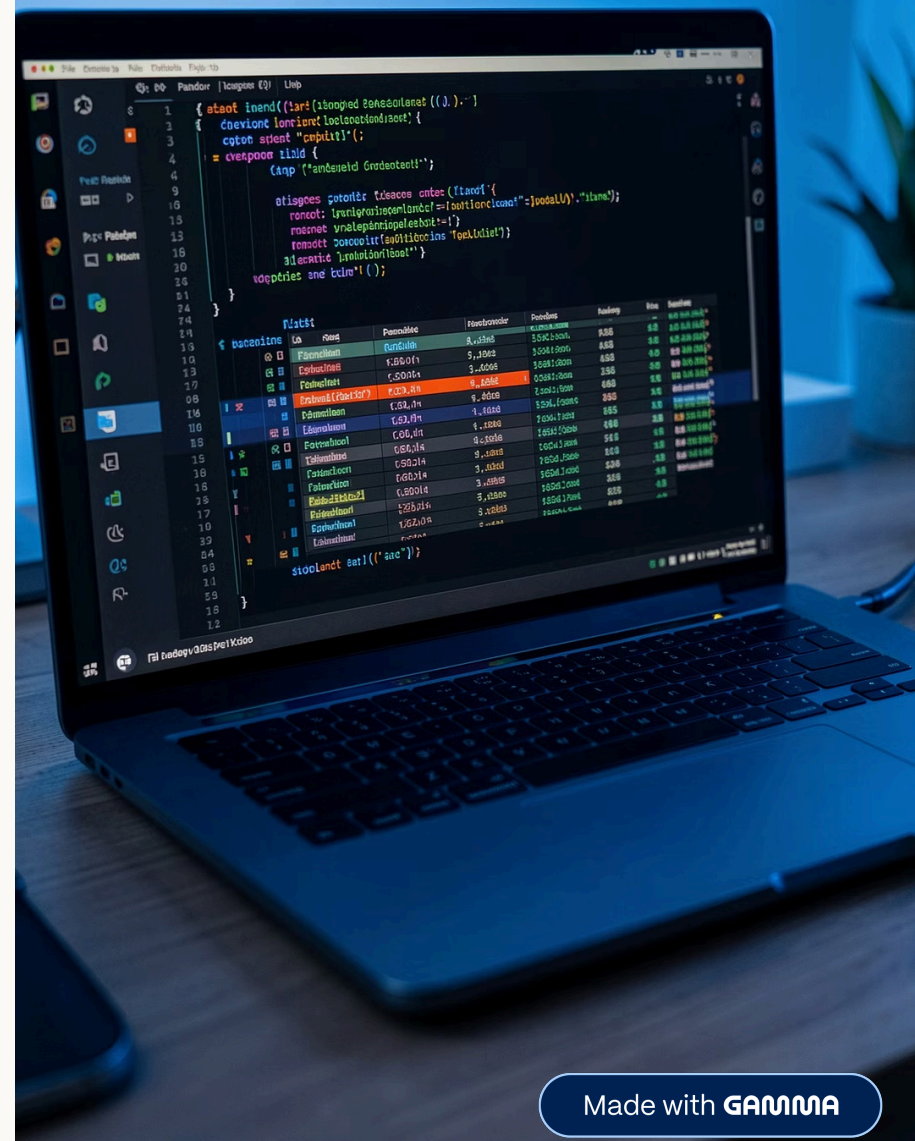
Imputed missing Review Ratings using median per category. Renamed all columns to **snake_case**.

Feature Engineering

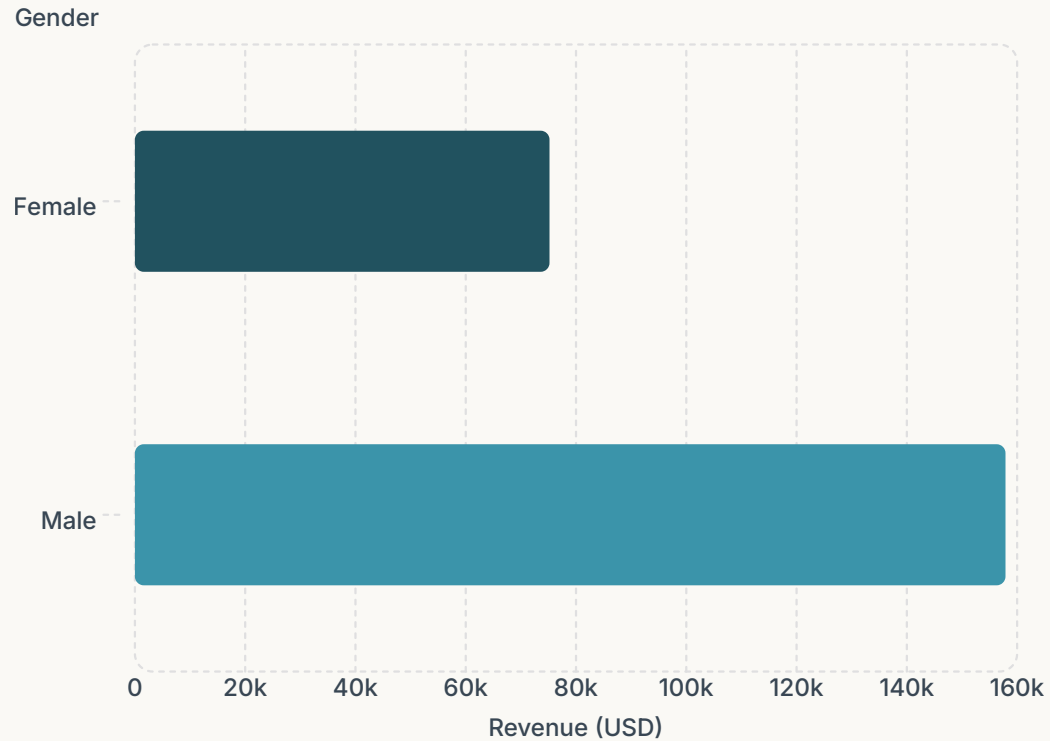
Created **age_group** (binned ages) and **purchase_frequency_days**.
Dropped redundant **promo_code_used** column.

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for SQL-based analysis.



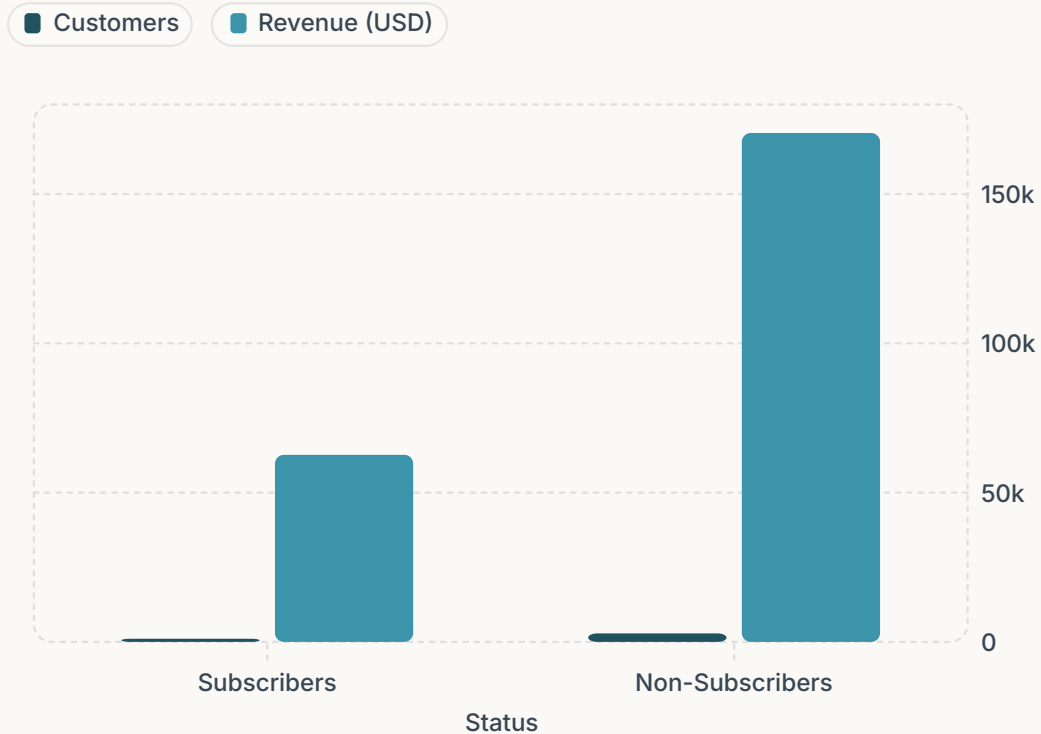
Revenue by Gender



Key Finding

Male customers generated **\$157,890** in revenue — more than double female customers at **\$75,191**. This represents a significant gender gap in total spend across the dataset.

Subscribers vs. Non-Subscribers



Insight

Non-subscribers dominate — **73% of customers** — generating **\$170,436** in revenue. Subscribers (27%) contributed \$62,645. Average spend is nearly identical (~\$59.50), suggesting subscription benefits may not yet be driving higher per-transaction value.

Top Products & Discount Dependence

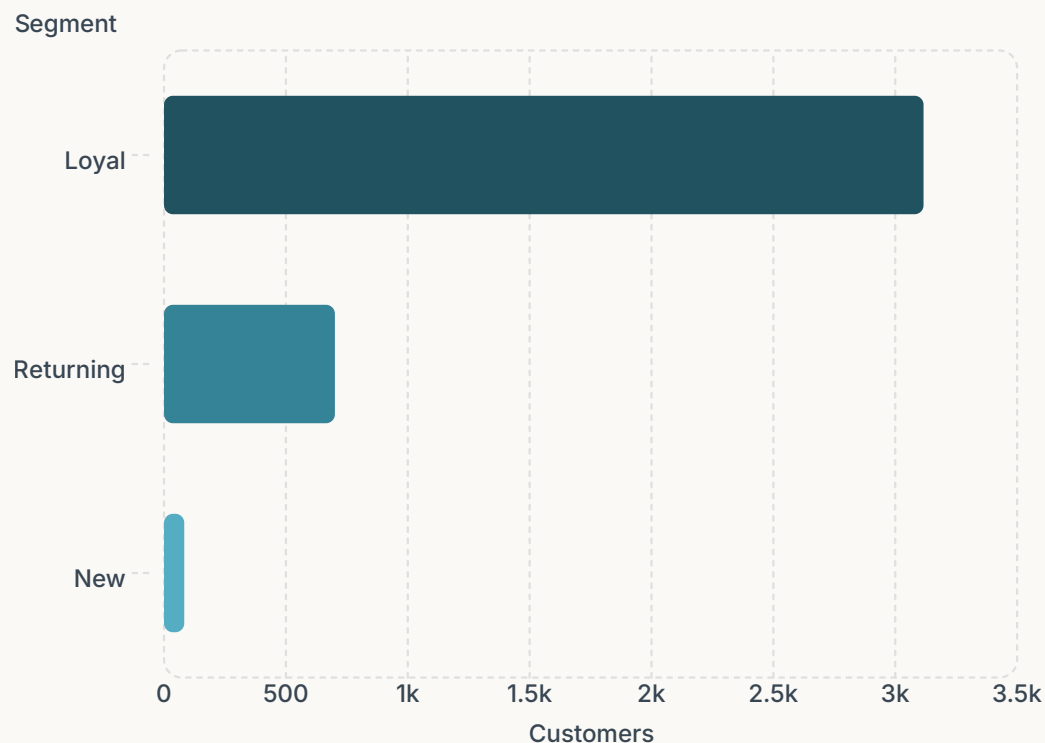
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rates

Product	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation



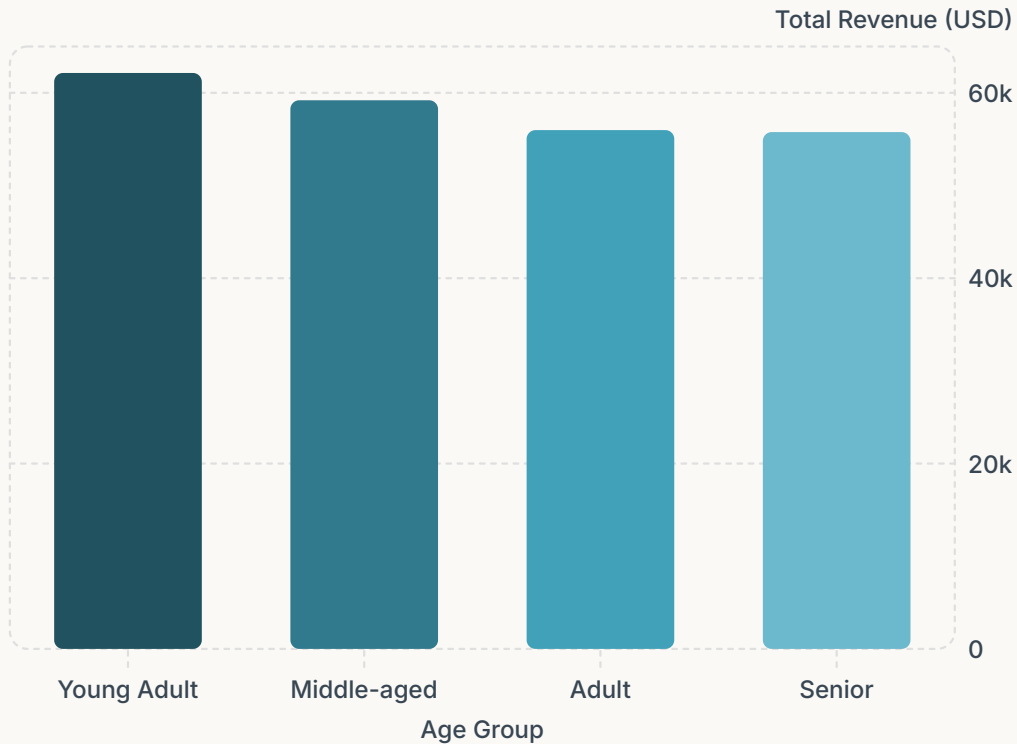
Segment Breakdown

Customers classified by purchase history into three segments:

- **Loyal (3,116)** — 80% of base, highest purchase frequency
- **Returning (701)** — 18%, moderate engagement
- **New (83)** — 2%, lowest purchase count

Of customers with **>5 purchases**, 958 are subscribers vs. 2,518 non-subscribers — repeat buyers are not yet converting to subscriptions at scale.

Revenue by Age Group



Age Group Insights

Young Adults lead in revenue at **\$62,143**, followed closely by Middle-aged (\$59,197). Revenue is relatively evenly distributed across all four age groups, with only a **\$6,380 spread** between highest and lowest.

Average purchase amounts are consistent across groups (~\$59.76 overall), suggesting age-based targeting should focus on **volume and frequency** rather than per-transaction value.

Power BI Dashboard



Key Metrics Displayed

- **3.9K customers** · **\$59.76** avg purchase · **3.75** avg rating
- **Revenue by Category:** Clothing ~\$100K, Accessories ~\$60K, Footwear ~\$30K, Outerwear ~\$15K
- **Sales by Category:** Clothing ~1,800, Accessories ~1,200, Footwear ~600, Outerwear ~300
- **Subscription:** 27% Yes · 73% No
- Interactive filters: Gender, Category, Shipping Type, Age Group



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to advance them into the "Loyal" segment.



Review Discount Policy

Balance sales uplift with margin control on heavily discounted products.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI.